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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Nexus
Selected Domain	Domain 1 - Aiversecity Future-Ready Workforce
Selected SDG Track (1–17)	4 - Quality Education; supporting SDGs 8 and 9 (project focus)
Project / Solution Name	NEXUS / AI Career Navigator
Name (Leader)	XIE YIFENG
Email & Phone (Leader)	ait2609068@xmu.edu.my | Phone: 1114401090
University Affiliation(s)	Xiamen University Malaysia (XMUM)
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

AI is changing work at the level of tasks, creating uncertainty about which skills students and workers should develop next. The World Economic Forum's 2025 employer survey expects 39% of existing skill sets to change or become outdated during 2025-2030 [2]. This is a global employer outlook, not an individual's probability of losing a job. In Malaysia, TalentCorp's Phase 1 impact study projects approximately 620,000 employees, about 18% within ten analysed sectors, to be significantly affected by AI, digitalisation and the green economy [1]. Impact does not mean redundancy, and the scope is not Malaysia's entire workforce.

For a university student or worker, a general career label is insufficient: they need to connect their circumstances, reported experience and concerns to occupational requirements and achievable next steps. A static interest test alone cannot show dated regional evidence or how a different learning budget changes a plan. NEXUS addresses this decision gap through source-linked exploration, explicit uncertainty and testable learning actions. No user-study or employment-outcome improvement has yet been established.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

NEXUS connects personal evidence to occupational requirements and a next action. Its starting-point questionnaire captures context, concerns, experience, goals, markets and AI practices. Twelve direction questions combine four common items with eight for a selected field, supporting ten directions. Self-report is never certified competence.

Local analysis compares answers with dated tasks and skills: 20 O*NET and seven ESCO records, plus geographically scoped regional software evidence. Route requirements reveal skills to investigate and inform personalised learning tasks. Broader directions retain bounded coverage.

The PR #14 career constellation and Skill DNA expose route relationships and requirements. What If compares hypothetical priorities, evidence and time budgets while preserving original answers. The existing 17-goal SDG view links actions to sustainability; this proposal focuses on education, decent work and innovation.

Optional consent-based RunPod interpretation uses validated evidence IDs; failure preserves the deterministic report. The model cannot add market facts or certify skills. English, Simplified Chinese and Malay support access. Replit hosts the prototype, while RunPod has documented GPU use. No sponsor-credit amount is claimed.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Frontend: native HTML, CSS and JavaScript ES modules; no React/Vue build requirement. Backend: Python 3.12, FastAPI, Pydantic, HTTPX and Uvicorn. Career analysis: local rules, evidence IDs and source-backed route requirements; advanced software comparisons keep unknown evidence explicit. Data: versioned O*NET/ESCO JSON plus bounded DOSM, World Bank, MyCOL, MASCO and BLS regional records. Optional legacy source radar retrieves O*NET pages/arXiv RSS. Optional AI: server-only RunPod vLLM, Qwen3-4B-Instruct-2507, schema/evidence validation. Storage: browser state/localStorage, JSON snapshots, temporary server caches and SQLite quota metadata; no profile database. Deployment: Replit FastAPI app, port 8000; GitHub source/CI and data-check code. RunPod and Replit were used; current optional-inference availability and scheduled execution are not claimed. See Appendix A.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We designed around a practical question: what should this person investigate next, given their concerns, reported experience and available time? The implementation uses structured, translated questions, adaptive follow-ups and a direct report instead of requiring a long free-text conversation. Evidence and unknowns remain visible so the user can challenge a recommendation.

Repository acceptance records document fictional-persona journeys, desktop/mobile/tablet layouts, language switching, scenario reset and failures involving sources, storage and optional AI. These are engineering checks, not user research or proof of career outcomes. The public homepage was accessible during this preparation; the full deployed commit remains unverified.

For Phase 2, we propose a small consented usability pilot using fictional profiles, independent review of source-to-action links, and completion of a practical learning task. Compare pre/post planning clarity and record denominators, failures and source dates. Retain the local report when external services fail. Deployment changes are outside this handoff; the existing Replit URL is the submitted prototype.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	4.4: relevant employment and entrepreneurship skills. 8.2: productivity through technological upgrading. 8.5: decent work (long-term contribution). 9.5: research/innovation capability (supporting contribution). These are proposed pathways, not measured national-indicator effects. [3]
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Published need: TalentCorp Phase 1 projects about 620,000 affected employees (18% within ten analysed sectors), not job losses. [1] Project evaluation baseline: 0 completed real-user impact studies established in inspected evidence. Planning-clarity and learning-completion baselines must be collected; no improvement rate has been measured.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Pilot KPI: change in planning-clarity rating (1-5); % participants able to identify one source-linked requirement and a feasible next task; % plans within declared weekly time; % reviewed recommendations with valid evidence/source/action links. Record sample size and missing responses. These are project proxies, not official SDG indicators.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Proposed Phase 2 acceptance targets: at least 80% of 10 pilot participants identify one sourced next action; 100% of reviewed plans respect the stated weekly budget; 100% of displayed citations resolve to approved records. Pilot and baseline remain unrun; targets are not results. No emissions or employment-rate reduction is claimed.

5. Eligibility & Readiness Checklist

Confirmed by team leader: all five members completed the required GenAI, Database and Cloud Deployment workshops; the project has a deployed SDG-linked prototype.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

No. The team leader confirms that no code, design or research from before the hackathon was used in this submission.

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Python/FastAPI stack	Code; see Technical Declaration inventory	PSF / MIT / BSD / MPL-2.0 / Apache-2.0 by component
O*NET 31.0; ESCO; regional sources	Public data / classification; selected adaptations	Source-specific terms; see Technical Declaration
Qwen3-4B-Instruct-2507	Pretrained model; optional RunPod inference	Apache-2.0; provider terms
NEXUS original interface artwork	Team-created for hackathon	Original team assets; project licence applies

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

No. The team leader confirms that no member has an employment, internship or consulting agreement giving an employer ownership of this submission.

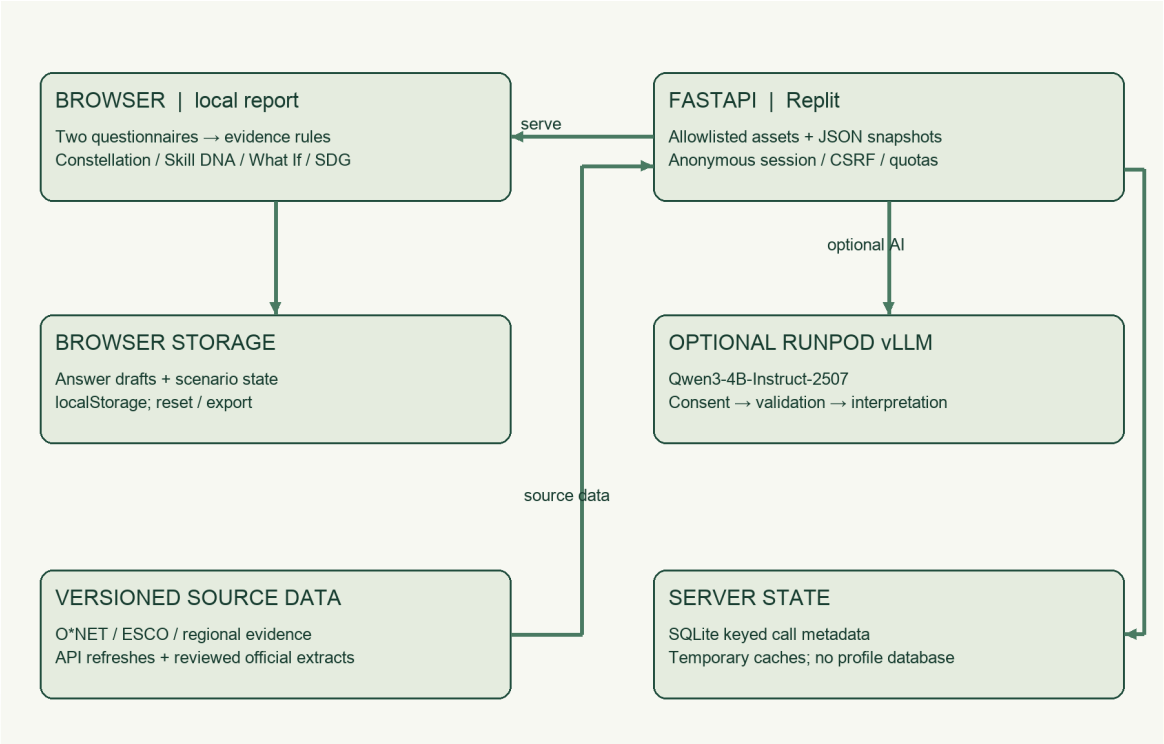
7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team’s knowledge and that the project will be original work created during the SDC 2026 hackathon period.

Signature: /s/ XIE YIFENG (authorised typed electronic signature)

Signing date: 25 October 2026 (MYT)

Appendix A System architecture



Solid connections describe implemented source components. Optional inference is a separate request after consent; it does not replace the base report. External data is versioned and scoped; reviewed document extracts are not presented as live APIs.

Frontend: native ES modules. Server: Python/FastAPI. Replit configuration: cloudrun target, scripts/replit_start.sh, local port 8000 exposed as port 80. Public URL availability is verified; exact production commit and current model availability are not.

Appendix B Sources and evidence boundaries

[1] TalentCorp, Impact Study of AI, Digital and Green Economy on the Malaysian Workforce, Phase 1 scope: 10 sectors, 949 roles; approximately 620,000 employees (18% in analysed sectors) expected to be significantly impacted. https://www.talentcorp.com.my/impact-study/

[2] World Economic Forum, Future of Jobs Report 2025. Employer survey outlook: 39% of existing skill sets expected to change or become outdated, 2025-2030. Global survey; not a Malaysian job-loss estimate. https://www.weforum.org/publications/the-future-of-jobs-report-2025/

[3] United Nations, SDG targets 4.4, 8.2, 8.5 and 9.5. Proposed contribution pathways; no national indicator changes claimed. https://sdgs.un.org/goals

[4] O*NET Database 31.0, USDOL/ETA. Selected occupation/task/skill records. CC BY 4.0; adaptations and translations by Nexus are not endorsed by USDOL/ETA. https://www.onetcenter.org/license_db.html

[5] European Commission ESCO classification. Selected occupation/skill relationships and reviewed contextual links; not job-demand statistics. https://esco.ec.europa.eu/en/use-esco

[6] Product source inspected from a read-only GitHub main ZIP on 25 October 2026 MYT. Main source and historical documentation can differ; no live commit identity is asserted. https://github.com/xieyifeng24/AI-Career-Navigator

[7] Live prototype opened successfully in Chrome on 25 October 2026 MYT. Exact deployment SHA and production optional-AI configuration remain unverified. https://ai-career-navigator-xyf20070924.replit.app/

Source files: prototype/combined/current-analysis.js; starting-point.js; intelligence.js; pages/; src/career/api.py, access.py, regional.py, regional_ai.py and sources.py; requirements.txt; .replit. The downloaded code was inspected without running or modifying product code. Product acceptance results quoted in repository documents are team records, not tests rerun during this handoff.

Knowledge snapshot: knowledge-20261024T081728Z-d5b9e134322f0749; verified 24 October 2026, 16:17 MYT. Regional snapshot: regional-5b92e0e0305b4a1a; checked 24 October 2026, 10:13 MYT. Snapshot times are not the statistical reference periods.